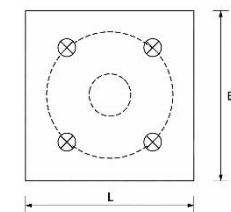
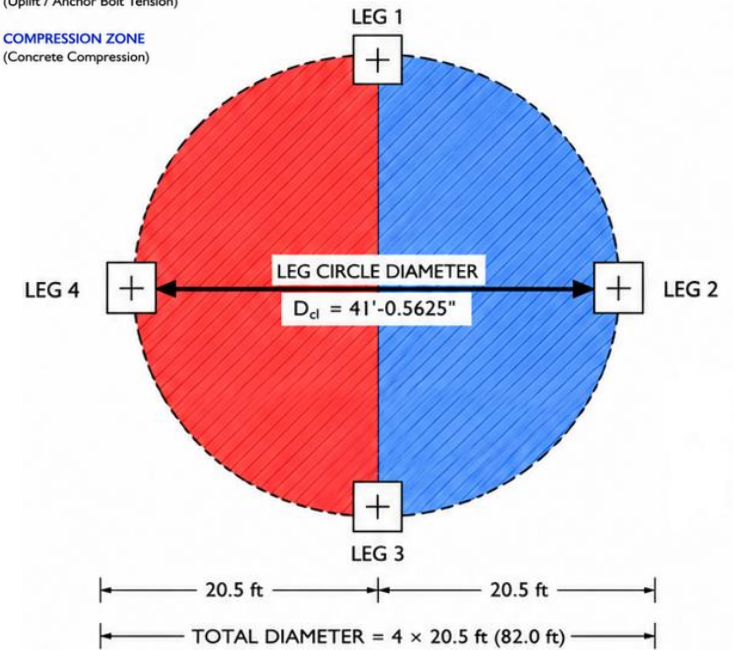
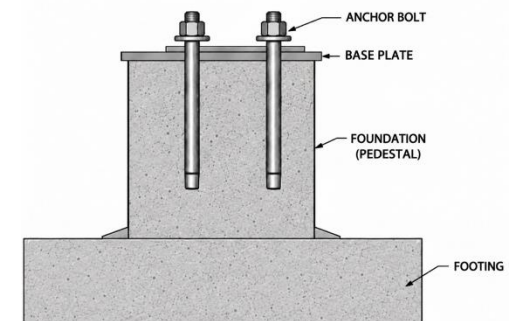


3: RINGWALL Foundation Design:

Design Parameter	Symbol	Value	Source
Number of legs	N_{leg}	4	User Input
Total factored axial load	P_u	1,609.756 kips	Main Tank Design (Predefined)
Governing factored overturning moment	M_u	5,673.437 kip-ft	Main Tank Design (Predefined)
Total factored horizontal shear	V_u	52.30 kips	Main Tank Design (Predefined)
Leg circle diameter	D_{cl}	41'-0.5625"	Main Tank Geometry (Predefined)
Pedestal size	$B \times L$	39 in $\times$ 39 in	User Input
Base plate size	D_{bp}	30 in	User Input
Concrete compressive strength	f'_c	4.0 ksi	User Input / ACI 318-19
Reinforcing steel yield strength	f_y	60 ksi	User Input / ACI 318-19
Anchor bolts	$N_b \times d_b$	4 $\times$ 1.75 in	Designed in Anchor Bolts segment of this software
Anchor embedment depth	h_{ef}	40 in	Designed in Anchor Bolts segment of this software
Pedestal clear height used	H_p	$\approx 7'-5" = 89$ in	User Input
Concrete unit weight	γ_c	150 pcf	User Input



SECTION CUT



Step 1: ACI 318-19 slenderness check

ACI 318-19 6.2.5.1 permits slenderness effects to be neglected for a column not braced against sidesway when:

For a square section:

$$r = \sqrt{\frac{I}{A}}$$

I = moment of inertia of the section

r = Radius of gyration

A = Sectional Area

For a 39 × 39in square:

For the 39 × 39in square:

$$\begin{aligned} I_x &= I_y = \frac{bh^3}{12} \\ I &= \frac{39(39)^3}{12} \\ I &= 192,571 \text{ in}^4 \end{aligned}$$

And:

$$\begin{aligned} A &= 39(39) \\ A &= 1,521 \text{ in}^2 \end{aligned}$$

Therefore:

$$r = \frac{39}{\sqrt{12}}$$

$$\mathbf{r = 11.26 \text{ inch}}$$

Check A — Pedestal classification

$$\frac{\ell_u}{b} \leq 3$$

$$2.28 < 3$$

Therefore, it qualifies geometrically as a pedestal.

Check B — Structural Slenderness

$$\frac{k\ell_u}{r} \leq 22$$

Using the conservative $k = 2.0$:

$$15.81 < 22$$

Therefore, second-order slenderness effects can be neglected.

Step 2: Gravity axial load per leg

For four equally spaced legs:

$$\begin{aligned} P_{gravity} &= \frac{P_u}{N_{leg}} \\ P_{gravity} &= \frac{1609.756}{4} \\ \mathbf{P_{gravity} = 402.44 \text{ kip/leg}} \end{aligned}$$

Step 3: Overturning force

From the anchor-bolt design:

$$r = 246.28 \text{ in}$$

and the overturning force calculation gives:

$$T = 207.3 \text{ kip}$$

with two legs on the tension side.

Therefore:

$$T_{leg} = \frac{207.3}{2}$$
$$T_{leg} = \mathbf{103.65 \text{ kip}}$$

This is the same tension-leg reaction established in the anchor-bolt design.

For pedestal design, use this force conservatively as the uplift/tension transfer demand.

Maximum Compression Pedestal Load

The compression-side leg receives the gravity reaction plus the overturning reaction:

$$P_{comp} = P_{u,leg} + T_{leg}$$
$$P_{u,ped} = 402.44 + 103.65$$

$$P_{u,ped} = \mathbf{506.09 \text{ kip}}$$

The pedestal self-weight can also be included.

For approximately 7'-5" pedestal height:

$$\gamma_c = 150 \text{ pcf}$$

$$W_p = (3.25)(3.25)(7.417)(0.150)$$
$$W_p \approx 11.75 \text{ kip}$$

Factored:

$$1.2W_p = 14.10 \text{ kip}$$

Therefore, conservatively:

$$P_{comp,total} = 506.09 + 14.10$$

$$P_{comp,total} \approx \mathbf{520.2 \text{ kip}}$$

Step 4: Pedestal Concrete Bearing Check

Base plate:

$$A_1 = \frac{\pi(30)^2}{4} = 706.86 \text{ in}^2$$

Pedestal:

$$A_2 = 39(39) = 1521 \text{ in}^2$$

Bearing enhancement factor:

$$\sqrt{\frac{A_2}{A_1}} = \sqrt{\frac{1521}{706.86}} = 1.467$$

Using the concrete bearing-strength relationship:

$$\phi B_n = \phi(0.85f'_c A_1) \sqrt{\frac{A_2}{A_1}}$$

Taking:

Transverse Pedestal Reinforcement: if

1. Spiral = 0.75
2. Ties = 0.65

$$\phi P_n = 0.65(0.85)(4)(706.86)(1.30)$$

$$\phi P_n \approx \mathbf{2292 \text{ kips}}$$

Demand:

$$P_u = 520.2 \text{ kip}$$

$$\mathbf{D/C = \frac{520.2}{2292}}$$

Therefore:

$$0.23 < 1.00 \dots \text{CONCRETE BEARING — PASS}$$

Demand/capacity ratio:..... $D/C \approx 0.20$

So the **39 × 39 in pedestal is more than adequate for bearing.**

Step 5: Pedestal longitudinal Reinforcement

The approved foundation drawing uses:

18 – #6 vertical bars

Area of one #6:

$$A_b = 0.44 \text{ in}^2$$

Therefore:

$$A_s = 18(0.44)$$

$$A_s = 7.92 \text{ in}^2$$

$$A_g = 39(39)$$

$$A_s = 1521 \text{ in}^2$$

Reinforcement ratio:

$$\rho = \frac{7.92}{39(39)}$$

$$\rho = 0.0053$$

or:..... $\rho = 0.53\%$

For the cast-in-place pedestal-to-foundation interface, ACI 318-19 requires reinforcement crossing the interface of at least:

$$A_{s,min} = 0.005A_g$$

ACI 318-19 Section 16.3.4.1 specifies the $0.005A_g$ minimum for connections between a cast-in-place column or pedestal and its foundation.

$$A_{s,min} = 0.001(1521)$$

$$A_{s,min} = 7.605 \text{ in}^2$$

Provided:

$$A_s = 7.92 \text{ in}^2$$

Therefore:

$$7.92 > 7.605 \dots \text{PASS}$$

Reinforcement ratio:

$$\rho_g = \frac{7.92}{1,521}$$

$$\rho_g = 0.00521 = 0.521\%$$

5:

$$A_g = 39(39) = 1521 \text{ in}^2$$

$$A_{s,min} = 0.005A_g$$

$$A_{s,min} = 0.005(1521) = \boxed{7.605 \text{ in}^2}$$

Provided:

$$A_s = 18(0.44) = \boxed{7.92 \text{ in}^2}$$

Therefore:

$$\frac{A_s}{A_{s,min}} = \frac{7.92}{7.605} = 1.041$$

$$A_s > A_{s,min} \dots \dots \text{OK}$$

Step 6: Axial Compression Capacity

Gross area:

$$A_g = 1,521 \text{ in}^2$$

Steel area:

$$A_s = 7.92 \text{ in}^2$$

Nominal concentric compression strength:

ACI 318-14 21.2.2, Table 21.2.2 — Strength reduction factor ϕ for moment, axial force, or combined moment and axial force.

Transverse reinforcement ϕ

1. Spirals conforming to §25.7.3 0.75
2. Other reinforcement / tied 0.65

$$\begin{aligned} P_o &= 0.85f'_c(A_g - A_s) + f_y A_s \\ P_o &= 0.85(4.0)(1,521 - 7.92) + 60(7.92) \\ P_o &= \mathbf{5,619.7 \text{ kips}} \end{aligned}$$

Using a conservative tied-member design strength factor and 0.80 maximum axial-strength factor:

$$\begin{aligned} \phi P_n &= 0.65(0.80)(5,619.7) \\ \phi P_n &= \mathbf{2,922.2 \text{ kips}} \end{aligned}$$

Then for **spiral reinforcement**:

$$\begin{aligned} P_n &= 0.85(5,619.7) \\ P_n &= 4,776.7 \text{ kips} \end{aligned}$$

Finally:

$$\begin{aligned} \phi P_n &= 0.75(4,776.7) \\ \phi P_n &= \mathbf{3,582.5 \text{ kips}} \end{aligned}$$

Demand is:

Therefore:

$$\phi P_n = 3,582.5 \text{ kips} > P_u = 520.2 \text{ kips}$$

and the utilization is:

$$\frac{P_u}{\phi P_n} = \frac{520.2}{3582.5} = 0.145$$

Demand:

$$P_u = 520.2 \text{ kips}$$

Therefore:

$$D/C = \frac{520.2}{2,922.2}$$
$$D/C = \mathbf{0.18}$$

Step 7: Pedestal Tensile Capacity Check

Governing tension-side pedestal demand:

$$T_u = \mathbf{103.65 \text{ kips}}$$

Provided longitudinal reinforcement:

$$A_s = 7.92 \text{ in}^2$$

Using:

$$\phi = 0.90$$

and:

$$f_y = 60 \text{ ksi}$$

The available steel tension strength is:

$$\begin{aligned}\phi T_n &= 0.90(7.92)(60) \\ \phi T_n &= \mathbf{427.68 \text{ kips}}\end{aligned}$$

Demand/capacity ratio:

$$\begin{aligned}D/C &= \frac{103.65}{427.68} \\ D/C &= \mathbf{0.24 \dots \dots \dots \text{PASS}}\end{aligned}$$

Therefore, the 18-#6 vertical reinforcement has adequate theoretical tensile capacity for the pedestal uplift-transfer demand.

Step 8: Pedestal P - M Interaction Check

8.1 Governing Axial Load and Moment

$$\begin{aligned}P_u &= 506.09 \text{ kip}, M_u = 97.0 \text{ kip-ft} \\ e &= \frac{M_u}{P_u} = \frac{97(12)}{506.09} = 2.30 \text{ in.}\end{aligned}$$

For the 39×39 in. pedestal:

$$d_t = 39 - \left(3 + 0.375 + \frac{0.75}{2} \right) = 35.25 \text{ in}$$

Because

$$2.30 < \frac{2}{3}(35.25) = 23.50 \text{ in.}$$

compression failure is initially assumed.

$$c_b = \frac{87}{87 + 60}(35.25) = 20.86 \text{ in.}$$

Iterating with $c > c_b$ using strain compatibility for the 18-#6 bars gives:

$$c = 41.21 \text{ in.}$$

and

$$a = \beta_1 c = 0.85(41.21) = 35.03 \text{ in.}$$

Concrete Compression Force

$$C_c = 0.85f'_c b a$$

$$C_c = 0.85(4)(39)(35.03) = 4,644.4 \text{ kip}$$

Its moment about the section centroid is:

$$M_c = C_c \left(\frac{39}{2} - \frac{35.03}{2} \right)$$

$$M_c = 769.1 \text{ kip-ft}$$

Reinforcing-Bar Moment

For each #6 bar:

$$\epsilon_s = 0.003 \frac{c - y_s}{c}$$

$$f_s = E_s \epsilon_s \leq f_y$$

The net force in each bar is:

$$F_s = (f_s - 0.85f'_c)A_b$$

Using the **18-#6 bar locations**, the reinforcing-bar moment about the section centroid is:

$$M_s = \sum F_s y_s$$
$$M_s = 175.6 \text{ kip-ft}$$

Total Nominal Moment

$$M_n = M_c + M_s$$
$$M_n = 769.1 + 175.6$$
$$M_n = 944.7 \text{ kip-ft}$$

The corresponding nominal axial strength is:

$$P_n = 4,929.0 \text{ kip}$$

For tied reinforcement, $\phi = 0.65$:

$$\phi M_n = 614.1 \text{ kip-ft}$$

Therefore:

$$\frac{M_u}{\phi M_n} = \frac{97.0}{614.1} = 0.158 < 1.0$$

Step 9: Horizontal Shear per Pedestal

Total tank shear:

$$V_u = 52.30 \text{ kip}$$

Assuming equal distribution among four legs:

$$V_{u,ped} = \frac{52.30}{4}$$

$$V_{u,ped} = 13.075 \text{ kip}$$

This is consistent with the anchor design, which subsequently distributes the pedestal shear among four anchor bolts and obtains 3.27 kip/bolt.

Step 9: Pedestal Tie Design

1. Tie diameter

25.7.2.2 Diameter of tie bar shall be at least (a) or (b):

(a) No. 3 enclosing No. 10 or smaller longitudinal bars

(b) No. 4 enclosing No. 11 or larger longitudinal bars or bundled longitudinal bars

ACI 318-19 25.7.2.2: For longitudinal bars #10 or smaller:

$$d_{bt} \geq \#3$$

Therefore:

$$d_{bt} = \#3$$

ACI 318-19 25.7.2.1 requires:

$$s \leq \min(16, d_b, 48d_{bt}, b_{\min})$$

Therefore:

$$16d_b = 16(1.00) = 16 \text{ in}$$

$$48d_{bt} = 48(0.375) = 18 \text{ in}$$

$$b_{\min} = 39 \text{ in}$$

Thus:

$$s_{\max} = \min(16, 18, 39)$$

$$s_{\max} = 16 \text{ in}$$

Adopt:

#3 closed ties @ 12 in. o.c.

This provides:

12<16 in

Final detail

Pedestal: 20-#8 vertical + #3 ties/crossties @ 12" o.c.

with the top 24 in. optionally detailed at 6 in. o.c. around the anchor/base-plate zone.

Step 10: Pedestal-to-Footing Development

ACI 318-19 25.4.2.

Tension development-length expression:

$$l_d = \frac{3}{40} \frac{f_y}{\lambda \sqrt{f'_c}} \frac{\psi_t \psi_e \psi_s \psi_g}{(c_b + K_{tr})/d_b} d_b$$

$$\lambda = 1.0$$

$$f_y = 60,000 \text{ psi}$$

$$f'_c = 4,000 \text{ psi}$$

$$\psi_t = \psi_e = \psi_s = \psi_g = 1.0$$

and conservatively:

$$\frac{c_b + K_{tr}}{d_b} = 2.5$$

Then:

$$l_d = \frac{3}{40} \frac{60,000}{\sqrt{4,000}} \frac{1.0}{2.5} (1.00)$$

$$l_d \approx 28.46 \text{ in}$$

The ACI minimum development length is:

$$l_d \geq 12 \text{ in}$$

Therefore:

$$l_d \approx 28.5 \text{ in}$$

Adopt:

$$l_d = 30 \text{ in minimum}$$

11.2 Development at the pedestal-footing interface

The pedestal longitudinal bars shall extend into the footing by at least:

$$30 \text{ in}$$

or the greater length required by the final ACI 318-19 development calculation based on the actual bar location and footing geometry.

The provided development length is therefore:

$$l_{prov} = 30 \text{ in}$$

and:

$$\frac{l_{prov}}{l_d} = \frac{30}{28.5} = 1.05$$

Thus:

$$l_{prov} \geq l_d \therefore \text{OK}$$

Factor	Condition	Value
ψ_t Casting position	More than 12 in. fresh concrete below horizontal bar	1.3
	Other conditions	1.0
ψ_e Epoxy/coating	Epoxy-coated, cover $< 3d_b$ or spacing $< 6d_b$	1.5
	Epoxy-coated, other conditions	1.2
	Uncoated or galvanized	1.0
ψ_s Bar size	#6 and smaller	0.8
	#7 and larger	1.0
ψ_g Steel grade	Grade 40 or 60	1.0
	Grade 80	1.15
	Grade 100	1.30

ACI 318-19 (25.4.2.5)